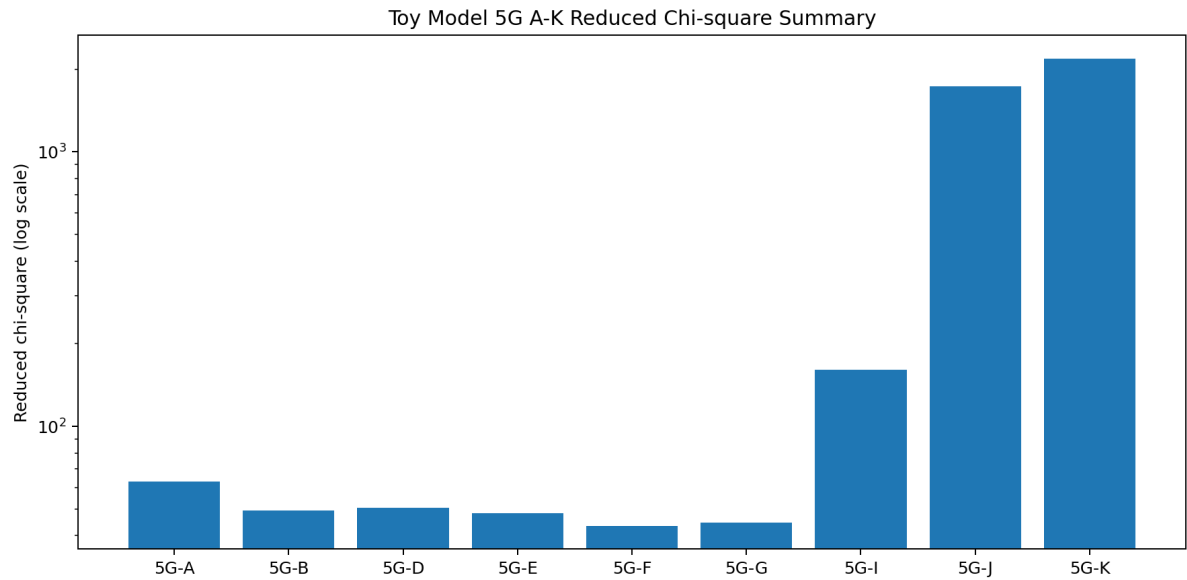


Toy Model 5G A-to-K Failure Analysis and Next Mathematical Baseline

Date: 2026.05.28
Status: Consolidation package. This is not a validation claim.

Executive conclusion. Toy Model 5G A through K did not produce a validated physical model. The sequence is valuable because it falsified several tempting shortcut paths: scalar acceleration thresholds, local compactness penalties, galaxy-wide mass scaling, simple nonlocal kernels, typed scheduling links, and Laplacian relaxation. The next mathematical object should be the inferred relational metric or connection, not another direct correction to velocity.



1. A-to-K Summary Table

Model	Method	Reduced chi-square	Conclusion
5G-A	One global A_R threshold; standard algebraic interpolation	63.181	Clean first falsification protocol. Massive spiral NGC2841 underfit; not validation.
5G-B	Global compactness response: $A_{\text{eff}}(R)=A_R[1+\beta Q(R)]$	49.512	Improved NGC2841 and DDO154, but worsened NGC3198.
5G-C	Residual anatomy; inner/transition/outer zone diagnostic	N/A	Showed NGC3198 worsened mainly in the inner zone, not transition zone.
5G-D	Saturated compactness response: $S=1+\beta Q/(1+Q)$	50.483	Saturation did not beat 5G-B. Useful failed branch.
5G-E	Global knot-tension: $A_{\text{eff},g} = A_R(M_b/M_{\text{ref}})^{\alpha}$	48.362	Geometrically cleaner; improves global chi-square vs B/D but still does not solve sample.
5G-F	Nonlocal spatial kernel: $\exp(- R_i-R_j /L_{\text{rel}})$	43.498	Best forward branch through F. Nonlocal web direction useful; NGC3198 still fails.
5G-G	Relational mass-space kernel: $\exp(- M_{\text{eff},i}-M_{\text{eff},j} /M_{\text{rel}})$	44.546	Mass-space ruler tested. Did not beat 5G-F; broad kernel acts like averaging.
5G-H	Inverse engine reconstruction; extracted $g_{\text{rel}} = g_{\text{obs}} - g_{\text{bar}}$	N/A	Found negative g_{rel} points in inner NGC3198; hidden engine cannot be a purely positive additive field.
5G-I	Typed dependency graph: FS/SS/FF and constrained SF	160.975 / 162.306	Failed badly. SF did not activate under gated rule. Typed matrix too crude.
5G-J	Relational tensegrity direct acceleration relaxation: $(I+\lambda L)^{-1} g_{\text{bar}}$	1736.029	Failed badly. Smoothing acceleration is not the engine.
5G-K	Variational potential relaxation: relax Φ , then gradient to g_{model}	2187.262	Failed badly. Simple quadratic Laplacian potential smoothing is not the engine.

2. RMS Residual Comparison

RMS residuals are in km/s. 5G-C was a diagnostic zone analysis and 5G-H was an inverse residual extraction, so they are not included as ordinary forward RMS models in this table.

Galaxy	5G-A	5G-B	5G-D	5G-E	5G-F	5G-G	5G-I	5G-J	5G-K
NGC2841	57.51	33.59	36.70	45.90	33.80	36.76	41.58	88.95	130.19
UGC00128	11.27	10.53	10.20	11.99	10.75	11.00	29.81	66.92	74.61
DDO154	4.01	3.52	3.46	2.55	3.07	3.09	6.51	21.90	25.97
NGC3198	10.06	13.96	14.41	12.82	14.90	14.76	34.27	36.58	55.85

Most important reading: 5G-F remains the best forward branch before the failed typed/Laplacian branches, but it still does not solve NGC3198. 5G-H then showed that the hidden residual includes negative points in inner NGC3198, which means the missing engine cannot be a simple positive additive acceleration field.

3. What Each Failure Taught

Scalar threshold failure. A single universal threshold alone cannot carry both massive compact spirals and diffuse dwarfs under strict global discipline.

Local compactness failure. Local density penalties can improve a massive spiral but damage inner regions where Newtonian baryonic behavior should dominate.

Knot mass failure. Galaxy-wide mass response is cleaner than local penalties but total mass alone is too crude; knot shape/topology matters.

Nonlocal kernel limit. The web intuition is valuable, but simple spatial or mass-space exponential kernels still do not define the true relational connection.

Inverse residual constraint. Negative g_{rel} points mean the hidden engine must redistribute, constrain, or modify geometry; it cannot simply add acceleration.

Scheduling-link failure. FS/SS/FF/SF time-scheduler analogues are conceptually useful but mathematically too crude for spatial equilibrium.

Laplacian relaxation failure. Smoothing acceleration or potential does not generate the required operational geometry. The metric/connection itself must be reconstructed.

4. Correct Next Mathematical Object: 5G-L

The next protocol should not try to fit the rotation curve directly. It should reconstruct the effective operational metric or connection implied by the observed curve, then test whether one universal baryonic-organization rule generates that geometry.

- Observed: $g_{\text{obs}}(R) = V_{\text{obs}}(R)^2 / R$
- Baryonic: $g_{\text{bar}}(R) = V_{\text{bar}}(R)^2 / R$
- Potential reconstruction: $\Phi_{\text{obs}}(R) = \int g_{\text{obs}}(R) dR$, $\Phi_{\text{bar}}(R) = \int g_{\text{bar}}(R) dR$
- Relational residual: $\Delta\Phi_{\text{rel}}(R) = \Phi_{\text{obs}}(R) - \Phi_{\text{bar}}(R)$
- Connection-like residual: $\Delta\Gamma_{\text{rel}}(R) = d[\Delta\Phi_{\text{rel}}]/dR$
- Test whether $\Delta\Phi_{\text{rel}}$ or $\Delta\Gamma_{\text{rel}}$ collapses onto a universal law of baryonic organization.

Success condition. One universal reconstruction/generation rule works across galaxies without per-galaxy halos, hidden rescue terms, or galaxy-specific metric laws.

Failure condition. Each galaxy requires its own metric law, its own edge-weight law, or hidden free parameters.

5. Publication Position

This package should be published as a disciplined falsification trail, not as a claim of success. It shows that the V4.2 framework is being tested against real data under strict guardrails and that failures are being preserved rather than hidden.

Recommended public wording: Toy Model 5G A-to-K tested whether dark-sector-like galaxy rotation behavior can emerge from relational capacity, scalar thresholds, nonlocal kernels, scheduling analogues, and variational spatial relaxation. These attempts did not validate the framework. They identified the next correct mathematical target: relational metric/connection reconstruction.